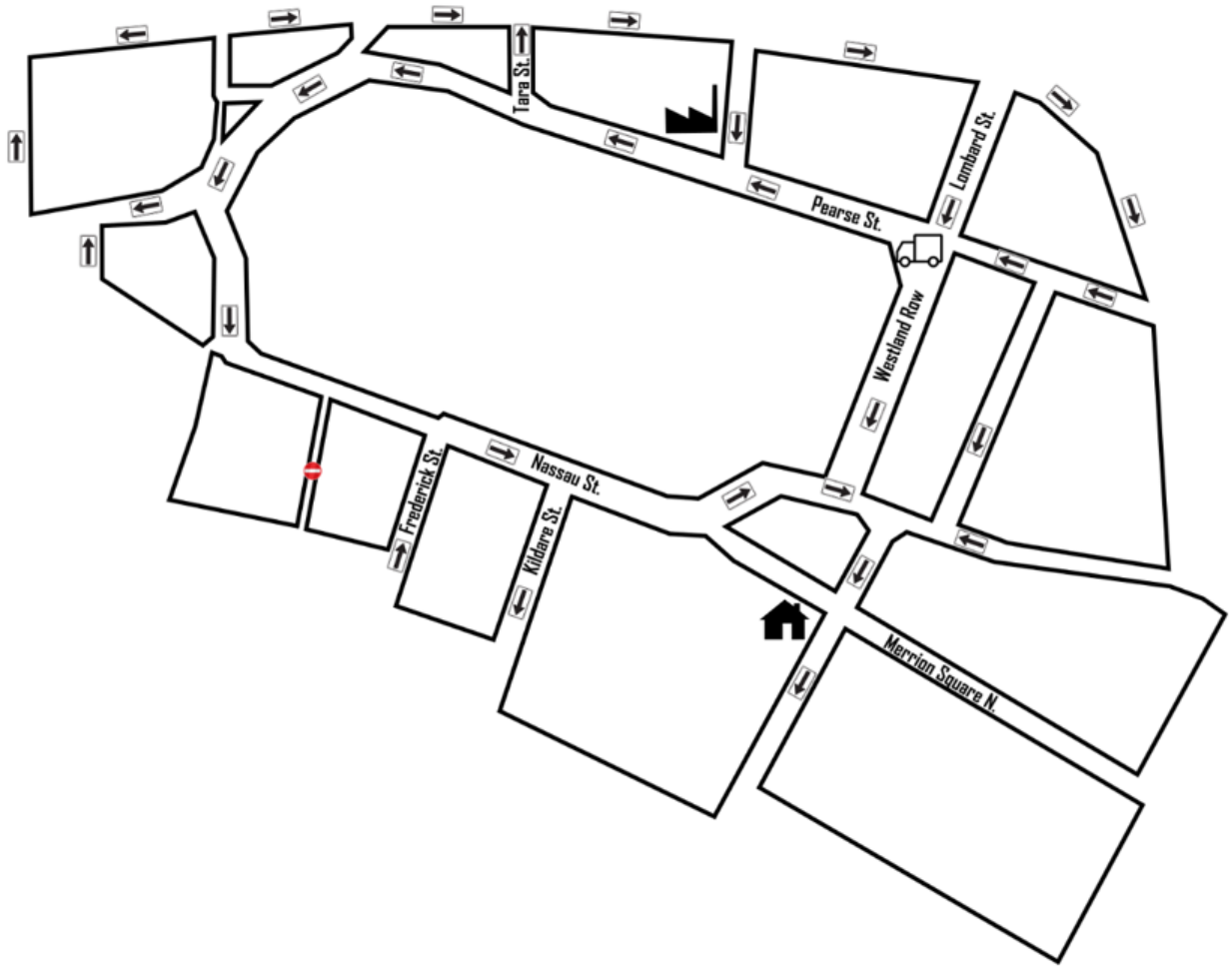



**Figure 14.15 An example of city map (Dublin’s city center), where the simplifications used to apply BFS for shortest paths wouldn’t be possible.**

## 14.4.1 Differences with BFS

Like BFS, Dijkstra's algorithm takes a graph and a source vertex as input (optionally a goal vertex as well), and computes the minimum distance from the source to the goal (or equivalently, with the same asymptotic running time, to all the other vertices in the graph). Differently than for BFS, though, in Dijkstra's algorithm the distance between two vertices is measured in terms of edges' weight. Consider